

Expert Interview Questions

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1 Introduction

Thank you for taking the time to participate in this interview and for reviewing the thesis overview beforehand.

The purpose of this discussion is to get your expert assessment of the proposed architecture, particularly its completeness, technical feasibility, practical applicability, and any limitations or gaps you identify.

As part of the expert validation, I will also ask you to complete a short evaluation form containing some background questions and structured ratings of the architecture. It should take about five minutes.

There are no right or wrong answers. I am mainly interested in your professional assessment, practical experience, and any areas where you think the architecture could be improved.

With your permission, I would like to record the interview so that I can transcribe and analyse it accurately. The recording and transcript will be pseudonymised, and names, company names, and other identifying information will not be included in the thesis analysis.

Do I have your permission to record the interview and use the transcribed responses for this thesis research?

2 Interview Questions

2.1 Architectural Completeness

1. Looking at the proposed architecture as a whole, do you think it contains the main functions and system areas required for a modern marine fire-suppression system?
2. Are there any important components, functions, interfaces, or architectural considerations that you believe are missing?

Optional prompts:

- Are the responsibilities of the different architectural areas sufficiently clear?
- Do you see any unnecessary complexity?
- Would you remove, combine, or separate any parts of the architecture?

2.2 Security Zones and Marine System Integration

3. From a marine system-integration perspective, does the proposed separation between safety-critical functions, supervisory functions, maintenance access, and external or customer

networks reflect how such a system should realistically be integrated onboard a vessel?

4. What should be considered when integrating this type of supplier system into the existing infrastructure of the shipyard, shipowner, or vessel?

Optional prompts:

- Are the proposed network boundaries realistic?
- Are the communication paths between the different system areas appropriate?
- Is the separation between safety functions and supervisory functions sufficient?
- How should customer or shipowner networks influence the architecture?
- Are supplier and customer responsibilities sufficiently clear?
- Should any interfaces be more tightly controlled or separated?

2.3 Cyber Resilience and Preservation of Essential Functions

5. If communication, supervisory functions, maintenance connectivity, or an external network connection were unavailable or compromised, do you think the proposed architecture would adequately preserve the essential fire-detection, alarm, control, and suppression functions? Why or why not?

Optional prompts:

- Are the proposed redundancy mechanisms sufficient?
- Is automatic failover realistic?
- Is degraded operation appropriate?
- Should the system be able to operate independently of supervisory or external networks?
- Do you identify any remaining single points of failure?
- Are there situations in which the proposed architecture could fail to preserve an essential safety function?

2.4 Cybersecurity and Maintainability

6. Do you consider the cybersecurity measures included in the proposed architecture appropriate and practically implementable for a marine fire-suppression system?

These measures include, for example:

- authentication and access control;
- network segmentation;
- restricted communication;
- controlled maintenance access;
- monitoring and logging;
- configuration and change management;
- protection of controller configurations;
- controlled remote connectivity; and
- recovery and restoration mechanisms.

7. Do any of these measures create potential difficulties for normal operation, commissioning, troubleshooting, service, or maintenance?

Optional prompts:

- Are the proposed access-control mechanisms practical?
- Could cybersecurity measures interfere with essential safety functions?
- Is controlled remote or maintenance access realistic?
- Are configuration and change-management requirements practical throughout the system lifecycle?

2.5 Fault and Cyber-Threat Assessment

8. The architecture has also been assessed using representative fault and cyber-threat scenarios. From your experience, do these represent realistic concerns for this type of system?
9. Are there any important failure modes or cybersecurity scenarios that you believe are missing?
10. At the conceptual or design level, do the proposed architectural safeguards appear sufficient to contain or reduce the effects of these scenarios? Why or why not?

Optional prompts:

- Are any of the assessed scenarios unrealistic or unlikely in practice?
- Are there additional scenarios that should be prioritised?
- Are there safeguards that would require further implementation-level verification or testing?

2.6 Industrial Applicability and Approval

11. If this framework were used to guide the design or review of a future marine fire-suppression system, would you consider it useful in an industrial project?
12. What would need to be changed, clarified, demonstrated, or documented before you would be comfortable using, reviewing, recommending, or approving such an architecture?

Optional prompts:

- Would the framework be useful during early system design?
- Could it support discussions with shipyards, shipowners, classification societies, or other stakeholders?
- What evidence or documentation would typically be required for acceptance or approval?
- Could the framework help reduce system-integration problems or design iterations?

2.7 Wrap-Up

13. Considering everything we have discussed, what do you see as the most important weakness or improvement opportunity in the proposed architecture?
14. Is there anything important regarding the architecture, marine system integration, cybersecurity, resilience, or practical implementation that I have not asked about but you think should be considered?

Thank you very much for your time and for sharing your experience.

Your feedback will be transcribed and analysed together with the other expert interviews. The findings will be used to assess the proposed architecture and, where appropriate, refine the architecture or identify areas requiring further implementation-level verification.